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What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 2: Two-Oscillator Entropy

Group-B

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1 Normal modes and the reduced state

We first test the entropy calculation on two coupled oscillators, where the ground state can be found exactly. In units $\hbar = 1$,

$$H = \frac{p_1^2 + p_2^2}{2} + \frac{1}{2}\boldsymbol{x}^T K \boldsymbol{x}, \qquad K = \begin{pmatrix} 1 & c \\ c & 2 \end{pmatrix}. \tag{1}$$

The eigenvalues are $(3 \pm \sqrt{1 + 4c^2})/2$. Both are positive when $|c| < \sqrt{2}$, including the required $c = 0.3$ and 0.6 cases. If $K = U\,\mathrm{diag}(\omega_j^2)U^T$, the change of variables $\boldsymbol{y} = U^T\boldsymbol{x}$ and $\boldsymbol{p}_y = U^T\boldsymbol{p}$ gives independent normal modes. Their ground-state variances are $1/(2\omega_j)$ and $\omega_j/2$. Transforming back gives

$$X = \langle \boldsymbol{x}\boldsymbol{x}^T \rangle = \tfrac{1}{2}K^{-1/2}, \qquad P = \langle \boldsymbol{p}\boldsymbol{p}^T \rangle = \tfrac{1}{2}K^{1/2}, \qquad \tfrac{1}{2}\langle \{x_i, p_j\} \rangle = 0. \tag{2}$$

Tracing out oscillator 2 retains $V_1 = \mathrm{diag}(X_{11}, P_{11})$. Its symplectic eigenvalue and entropy are

$$\nu = \sqrt{\det V_1}, \qquad s(\nu) = (\nu + \tfrac{1}{2})\ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2})\ln(\nu - \tfrac{1}{2}). \tag{3}$$

The convention $0 \ln 0 = 0$ gives zero entropy at $\nu = 1/2$. At nonzero coupling, the normal modes involve both original oscillators. The joint ground state remains pure, while the reduced oscillator can be mixed. Entropy is measured in nats.

Coupling c	u	Entropy (nats)
0	0.5	0
0.3	0.502816542	0.0193599304
0.6	0.512481339	0.0672711138

2 Numerical checks and interpretation

The diagonalisation reproduces $\nu = 0.5028165419863$ and $s = 0.0193599304236$ at $c = 0.3$. At $c = 0$ the entropy vanishes. At $c = 0.6$ it increases to 0.06727. A separate check uses the ground-state Gaussian wavefunction kernel $A = K^{1/2}$. Direct Gaussian integration gives

$$\mathrm{Tr}\rho_1^2 = \sqrt{\frac{\det A}{A_{11}A_{22}}} = \frac{1}{2\nu}, \tag{4}$$

which agrees with the covariance result. A local rescaling $x_1 \mapsto bx_1$, $p_1 \mapsto p_1/b$ changes the individual variances but leaves their product and s unchanged. The largest determinant change over $b = 0.1, 0.5, 1, 2, 10$ is 5.551×10^{-17}. Thus squeezing a decoupled oscillator does not create inter-oscillator entanglement.

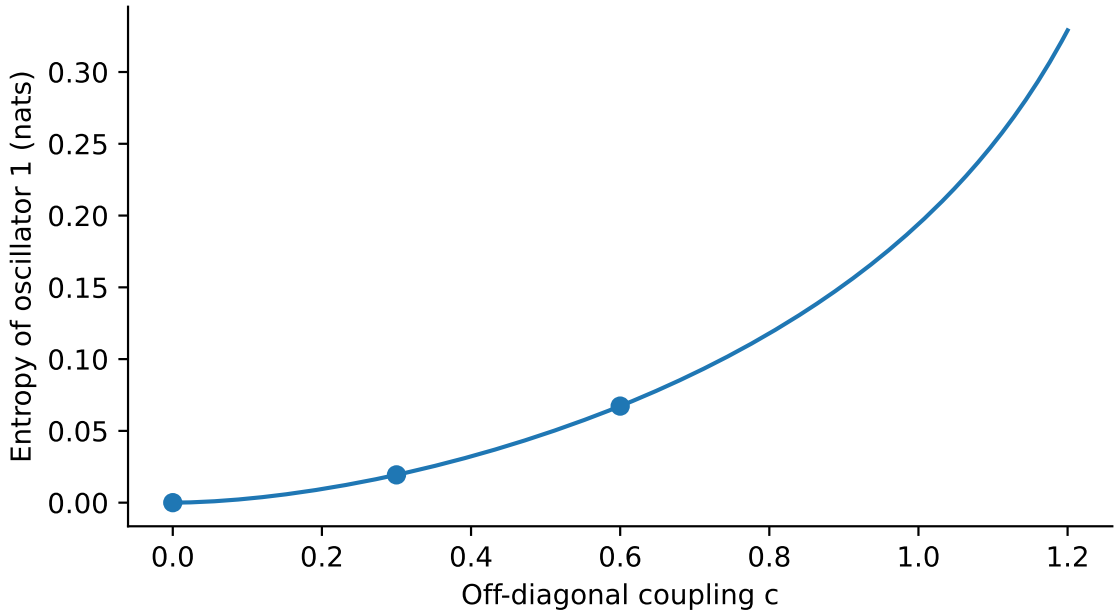


Figure 1: Entropy of one oscillator versus coupling. The marked points include the decoupled case and the two assigned benchmarks.

For real fields, $q_{-\boldsymbol{k}} = q_{\boldsymbol{k}}^\dagger$. A generic nonzero pair $\pm\boldsymbol{k}$ splits into two identical real cosine/sine sectors. Each contains one light and one spectator oscillator. We report $s(\nu_q)$ per real sector; the complete pair contributes $2s(\nu_q)$.

References

[1] C. Weedbrook et al., *Gaussian Quantum Information*, Rev. Mod. Phys. 84, 621 (2012), arXiv:1110.3234.

Code and supporting data: step2 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Normal-mode derivation and benchmark review.
Samarth Bhatnagar	Reduced-state covariance and rescaling checks.
Mayank Sinha	Numerical diagonalisation, purity cross-check and figure.